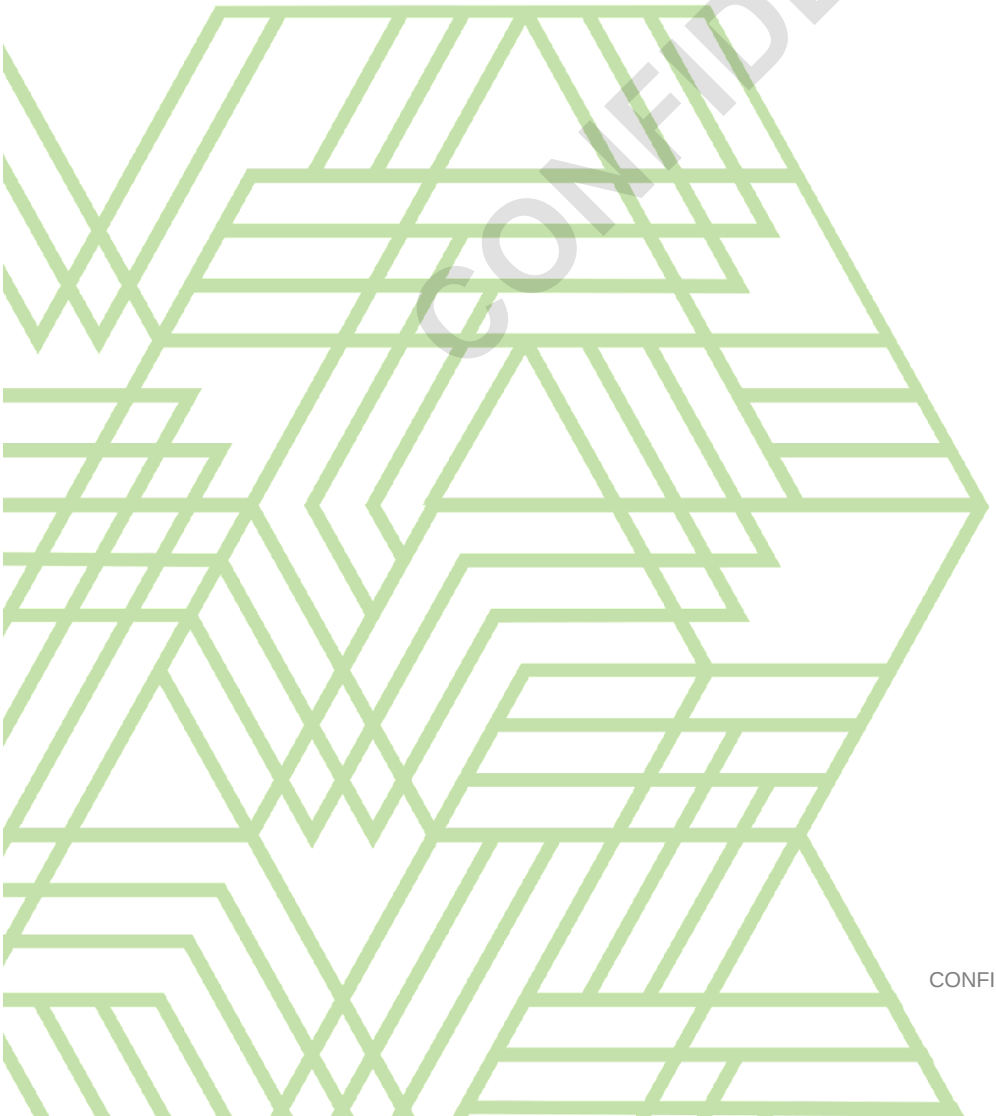
ATM33/e Series Pulse Width Modulation

Application Note

SUMMARY: This document describes the Pulse Width Modulation (PWM) hardware operation of the ATM33/e Wireless SoC Series.

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Table of Contents

1. Overview 3

2. PWM Capabilities 3

2.1 Continuous Mode 3

2.2 Counter Mode 3

2.3 IR Mode 4

2.4 IR FIFO Mode 4

3. Further References 5

Revision History 5

List of Figures

Figure 1 - Continuous Mode Frame
Figure 2 - Counter Mode Frame
Figure 3 - IR Mode Frame

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1. Overview

The ATM33/e (a.k.a ATM33/ATM33e) consists of:

- ATM33e series
  - ATM3330e 7x7 mm package
- ATM33 series
  - ATM3330 7x7 mm package
  - ATM3325 5x5 mm package

This application note describes PWM hardware operation of the ATM33/e Wireless SoC Series.

Please refer to the ATM_pwm example as well as the IR driver in the SDK for further reference.

2. PWM Capabilities

- PWM core supports upto 8 channels.
- On each channel, the application can control duty cycle by specifying high and low time duration. High/Low duration times are set in clk_mpc (16 MHz) clock cycles.
- Each channel can support freq range from 122 Hz to 8 MHz.
- Each PWM channel can operate in the 4 modes (Continuous/Counter/IR/IR_FIFO)

2.1 Continuous Mode

In continuous mode PWM will continuously generate output based on programmed duty cycle duration. If duty cycle duration is changed during runtime, updated parameters take effect on the next frame.

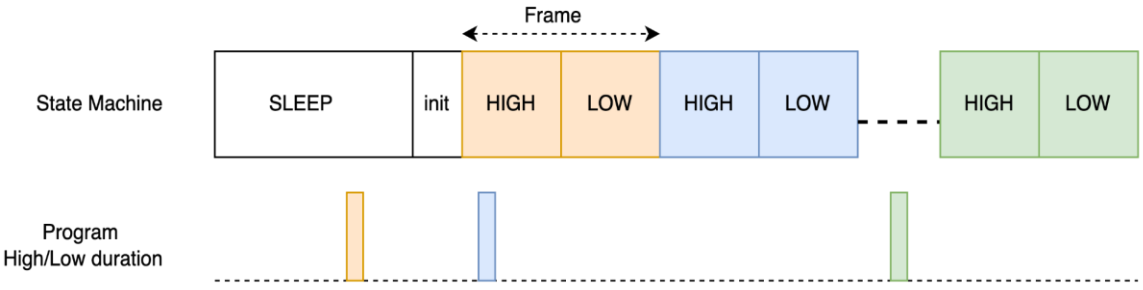


Figure 1 - Continuous Mode Frame

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2.2 Counter Mode

In Counter mode, software programs the high and low duration along with the number of frames to be generated.

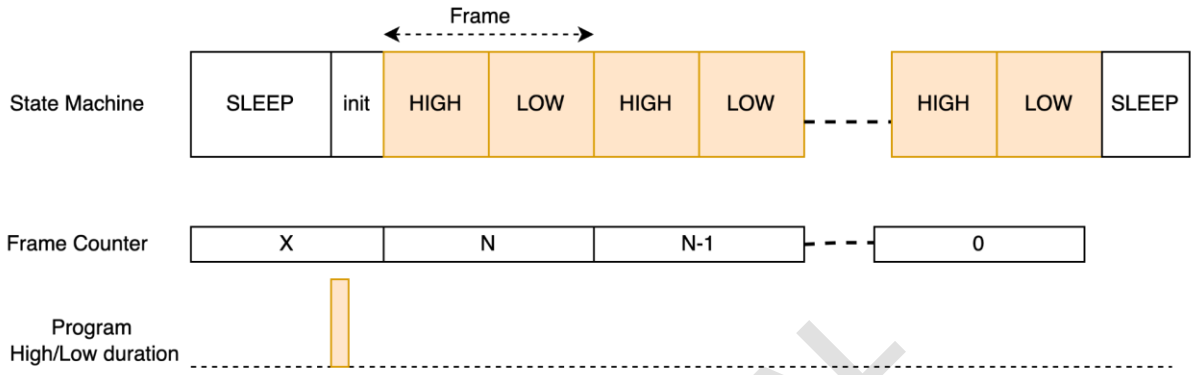


Figure 2 - Counter Mode Frame

2.3 IR Mode

In IR mode, software can keep on enqueuing new freq. sequences. Hardware on completion of the previous sequence starts a new one right afterwards.

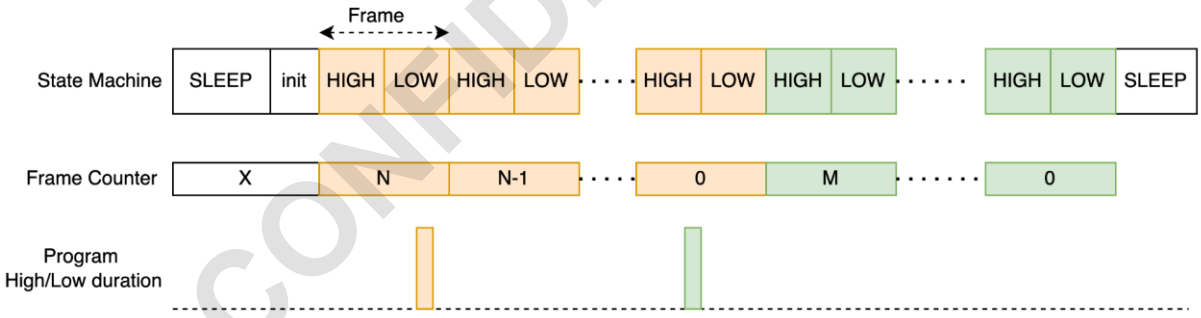


Figure 3 - IR Mode Frame

2.4 IR FIFO Mode

In IR_FIFO_MODE, the timing is identical to IR_MODE. The only difference is how the timing parameters are controlled.

In this mode, two static parameters [`fifo_carrier1_dur`, `fifo_carrier2_dur`] are used to configure carrier frequency and duty cycle (**hi_dur[15:0]**, **lo_dur[15:0**]), so users can configure two carrier frequencies.

`Frame_count`, `freq_sel`, and `IR_region_sel` come from the FIFO buffer instead of the memory-mapped register file.

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FIFO can hold upto maximum of 16 entries. FIFO configuration (via **FIFO_CFG** register) allows Software to program low water mark which triggers interrupt when FIFO is emptied. Software on receiving interrupt can push more commands into FIFO.

To push an item into the FIFO, DMA or MCU writes a new value to the register **fifo_data[15:0]** which has the following mapping:

[15] freq_sel: selects carrier frequency

0: use carrier frequency defined by fifo_carrier1_dur register

1: use carrier frequency defined by fifo_carrier2_dur register

[14] IR_region_sel: selects between mark and space IR regions

0: output lo for the space region in IR.

space region frame: `hi_dur`=0, lo_dur= hi+lo dur of selected carrier frequency, for example, if carrier1_dur is selected, lo_dur during space region is the sum of carrier1 hi_dur+lo_dur. Max lo_dur is limited to 16-bit, in other words, the sum of the selected carrier hi+lo_dur shouldn't exceed 16-bit.

1: output value defined by hi/lo duration for the mark region in IRi_dur

[13:0] frame_count

***Note**: Only one PWM channel can operate in IR_FIFO mode.*

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3. Further References

Users are encouraged to read the following documents to gain a more in-depth understanding of the architecture and software development on Atmosic SDK.

- ATM33/e Series SDK User Reference Manual.

Revision History

Date	Version	Description
August 16, 2022	0.50	Initial version created.

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